Assignment # 5

AI Application Architecture

[AI Workspace – System Architecture & Request Flow]

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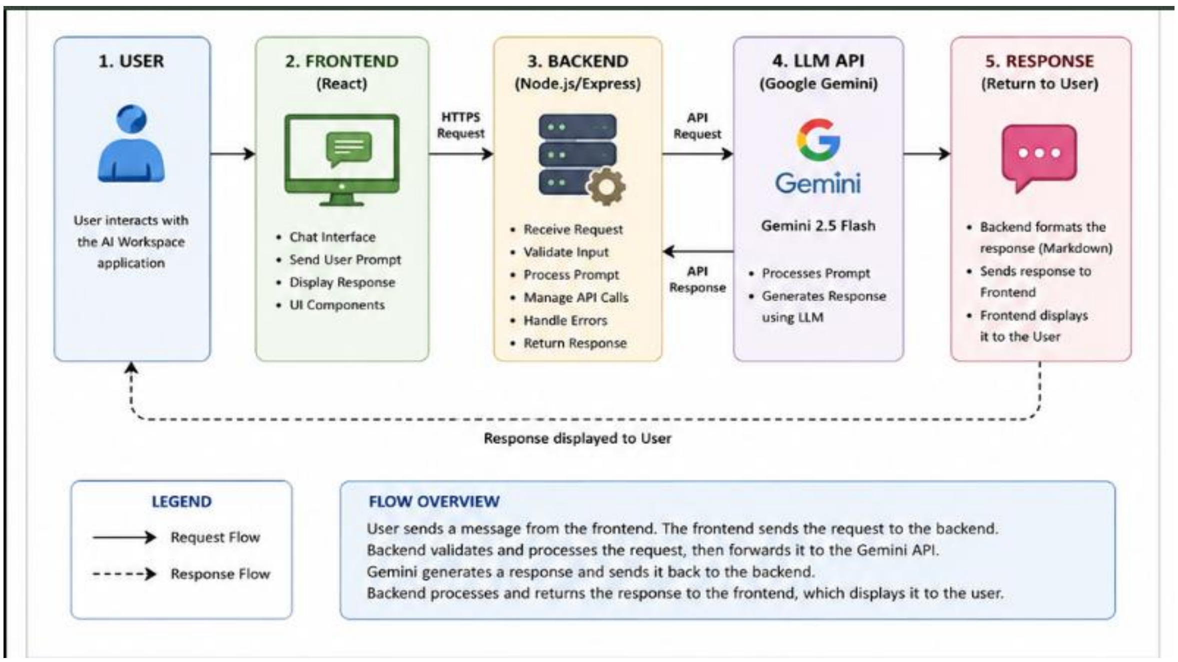
Introduction:

Artificial Intelligence applications are built using multiple interconnected components that work together to process user requests and generate intelligent responses. An AI application architecture diagram provides a visual representation of these components and illustrates how information flows between users, frontend interfaces, backend services, Large Language Models (LLMs), databases, and external APIs.

A well-designed architecture improves scalability, maintainability, security, and overall system performance. It also helps developers understand the responsibilities of each component and simplifies future enhancements.

Architecture Diagram:



This report presents the architecture of the AI Workspace application developed during the AI Engineering Internship. The architecture demonstrates the interaction between the user interface, application logic, prompt management, AI models, and supporting services.

Explanation:

1. Request Flow:

When a user enters a message in the AI Workspace, the frontend sends the request to the backend. The backend validates the request and forwards it to the Google Gemini API. The AI processes the prompt and returns a response.

2. Response Flow:

The Gemini API sends the generated response back to the backend. The backend formats the response (including Markdown if needed) and returns it to the frontend, where it is displayed in the chat interface.

3. Error Handling:

The application handles common errors including:

- Invalid API key
- Empty user prompt
- Network or connection failures
- API timeout
- Model unavailable

Appropriate error messages are shown to the user to improve usability.

4. API Integration:

The application integrates with the Google Gemini API using an API key stored securely. The frontend never communicates directly with the AI model; all requests pass through the backend to improve security and manage API communication.

Conclusion:

The AI Workspace follows a standard AI application architecture consisting of a frontend, backend, LLM API, and response pipeline. This architecture ensures scalability, security, and efficient communication between the user and the AI model.